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MH1200 Linear Algebra I

Hans Gabriel D. De Vera
Nanyang Technological University
hansgabr001@e.ntu.edu.sg

1. Subspaces Induced by Homogenous Systems

Theorem Solutions of Homogenous Systems — Theorem 1

The solution set of a homogenous system forms a subspace.

Proof Solutions of Homogenous Systems — Proof 2

All homogenous systems can be written in the form:

$$C\vec{x} = \vec{0}$$

Where C is a matrix containing the coefficients of the equations in the system.

Clearly, $\vec{x} = \vec{0}$ is a solution since any matrix times the zero vector results in the zero vector.

Further, if $\vec{x}$ and $\vec{y}$ are solutions, then we have:

$$\vec{0} = \vec{0} + \vec{0} = C\vec{x} + C\vec{y} = C(\vec{x} + \vec{y})$$

So indeed, $\vec{x} + \vec{y}$ is also a solution.

Finally, suppose $\vec{x}$ is a solution. For any $k \in \mathbb{R}$, we have:

$$\vec{0} = C\vec{x} = kC\vec{x} = C(k\vec{x})$$

Thus, the solution set of the homogenous system satisfies all axioms of a subspace and is thus a subspace, as needed. ■

2. Linear Combination

Definition Linear Combination — Definition 3

$\vec{w}$ is a linear combination of vectors $\vec{v_i}$ iff there exist scalars c_i such that:

$$\vec{w} = \sum_{i=1}^{n} c_i \vec{v_i}$$

Theorem Special Vectors in a Linear Combination — Theorem 4

The vectors in a linear combination are themselves linear combinations of the whole set.

Additionally, the zero vector is a linear combination of the vectors in the set.

Now, how do we decide if this is satisfiable?

$$\vec{w} = \sum_{i=1}^{n} c_i \vec{v_i}$$

We can construct the following augmented matrix:

$$\left(\begin{array}{cccc|c} \vdots & \vdots & \vdots & \vdots & \vdots \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \vec{v_1} & \vec{v_2} & \vec{v_3} & \vec{v_n} & \vec{w} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{array}\right)$$

If this augmented matrix is consistent, then $\vec{w}$ is a linear combination of the vectors $\vec{v_i}$.

Furthermore, if the augmented matrix has no free parameters, then the set of vectors are pairwise linearly independent and thus form a basis.

Definition Span

Definition 5

Suppose $B \subset \mathbb{R}^n$ is a subset of vectors with $|B| = m$. Then, the span of B is defined as:

$$\text{span}(B) := \left\{ \sum_{i=1}^{m} c_i \vec{b_i} \in \mathbb{R}^n \mid c_i \in \mathbb{R}, b_i \in B \right\}$$

Theorem Subspaces Induced by Span

Theorem 6

Consider a set of vectors in $\mathbb{R}^n$, then the set of all possible linear combinations of this set forms a subspace.

That is, $\text{span}(B)$ is a subspace of $\mathbb{R}^n$.